Python Programming — Week 4

String Methods Assignment

Solve all questions using only the methods taught in class

Name: ______________________________	**Date:** ____________________	**Score:** _____ / 100

Instructions: Read each question carefully. Write your Python code in the space provided. You may use any string methods covered in class. Pay attention to edge cases. Marks are awarded for correct output and clean, readable code.

A Main Project — Interactive String Menu [40 marks]

Build a **menu-driven program** that continuously accepts a string from the user and lets them perform various string operations. The program must keep running until the user chooses to exit.

Sample Run: Enter a string: hello world python ===== STRING MENU ===== 1. Strip Spaces 2. Find a Substring 3. Capitalize String 4. Convert to Uppercase 5. Convert to Lowercase 6. Convert to Title Case 7. Replace a Word 8. Count Occurrences 9. Check Starts With 10. Check Ends With 11. Split String 12. Exit Enter your choice: 3 Result: Hello world python

Your program must implement **at least 10 menu options** including the ones above. The first 5 options shown (Strip, Find, Capitalize, Upper, Lower) are mandatory. Add at least 5 more from the list below:

- Title Case conversion
- Replace a word / substring
- Count occurrences of a substring
- Check if string starts with a given prefix
- Check if string ends with a given suffix
- Split the string on a delimiter chosen by the user
- Reverse the string
- Check if the string is a palindrome

- Remove all vowels from the string
- Count the number of words

Write your code here:

B Short Answer Questions [30 marks]

Q1 Predict the Output

What will the following code print? Write the output exactly.

Expected Output

```
name = "   Hello World   "
print(name.strip().lower())
print(name.lstrip().rstrip())
print(name.strip().title())
```

Answer / Output:

Q2 Find the Bug

The code below has a bug. Identify and fix it. Explain why it was wrong.

Expected Output

```
fruits = "apple,banana,mango"
fruits_list = fruits.split(",,")
print(fruits_list)
# Expected: ["apple", "banana", "mango"]
# Actual output is wrong — fix the bug!
```

Answer / Output:

Q3 Fill in the Blank

Complete the code so that it prints True if the email is valid (contains '@' and ends with '.com').

Expected Output

```
email = "user@example.com"
is_valid = email.______("@") != -1 and email.______(".com")
print(is_valid)   # → True
```

Answer / Output:

Q4 Write the Code

Write a single line of Python that takes the string below, strips spaces, converts to lowercase, and replaces every 'python' with 'programming'.

Expected Output

```
text = "  I love PYTHON and Python is great  "
# Your one-liner here:
result = _______________________________________________
# Expected: "i love programming and programming is great"
```

Answer / Output:

Q5 Trace the Code

Trace through the code step by step and write the value of 'result' after each line.

Expected Output

```
sentence = "  the quick brown fox  "
result = sentence.strip()          # result = ?
result = result.title()            # result = ?
result = result.replace("The","A") # result = ?
words  = result.split()            # words  = ?
final  = "-".join(words)           # final  = ?
```

Answer / Output:

C Applied Problems — Real-World Scenarios [30 marks]

Q1 Email Validator [10 marks]

Write a function `validate_email(email)` that returns **True** if the email is valid, **False** otherwise. A valid email must: (a) contain exactly one '@', (b) end with '.com', '.org', or '.in', and (c) not have any spaces.

Test your function with at least 3 different inputs — one valid, two invalid.

Write your function and test cases here:

Q2 CSV Data Parser [10 marks]

You receive student data as a single comma-separated string. Write code to:

- Split the string into individual records
- For each record, extract the student name (first field) and print it in Title Case
- Count and print the total number of students

Expected Output

```
data = "tanishq tyagi,22,delhi|priya sharma,20,mumbai|rahul verma,21,pune"
# Hint: first split on "|", then split each record on ","
# Expected output:
# Tanishq Tyagi
# Priya Sharma
# Rahul Verma
# Total students: 3
```

Answer / Output:

Q3 Username Generator [10 marks]

Write a function `generate_username(full_name)` that creates a username from a full name following these rules:

- Remove all leading and trailing spaces
- Convert to lowercase
- Replace spaces between words with an underscore (_)
- If the username is longer than 15 characters, truncate it to 15

Expected Output

```
# Example:
# generate_username("  Tanishq Tyagi  ")  →  "tanishq_tyagi"
# generate_username("  Alexander Hamilton  ")  →  "alexander_hami"  (15 chars)
```

Write your function here:

★ Bonus Challenge [+10 marks]

Palindrome Checker with Cleaning: Write a function `is_palindrome(text)` that returns True if the text is a palindrome. The function must ignore spaces, punctuation differences, and casing. For example, *"A man a plan a canal Panama"* should return True.

Hint: Use strip(), lower(), replace(), and string slicing with [::-1].

Answer / Output:

Submission Guidelines: Submit your .py file named `YourName_StringAssignment.py` • Ensure the code runs without errors • Add comments explaining your logic • Due date will be announced in class